Sirui Tao

Researcher, Engineer, and Designer

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I am a Cognitive Science Ph.D. student at UC San Diego studying HCI, design, and human-centered AI. My work asks how interfaces can help people compare options, reason through tradeoffs, and decide what is worth making.

Education

University of California San Diego - La Jolla, CA, USA

2025-08 - 2030-06 **Doctor of Philosophy - Cognitive Science**
Research in Human-computer-interaction & Design. Advised by Steven P. Dow.

2023-09 - 2025-03 **Master of Science - Computer Science and Engineering**
Research in Human-computer-interaction & Graphics. Advised by Steven P. Dow and Tzu-Mao Li.

2019-09 - 2023-03 **Bachelor of Science - Data Science & Probability and Statistics**
Research in Intuitive Physics. Advised by Judith E. Fan.

Publications (& Rejections)

[P5] ~~***Manuscript under review.***~~
Sirui Tao and Steven P. Dow.
Under review (2026). **[Rejected at CHI26 & UIST 2026] :-<**

[W1] ***What Happened and Why? Trace-Guided Micro-Episodes with Elicited User Explanations for Product Iteration***
Sirui Tao, William P. McCarthy, Steven P. Dow.
Conference on Human Factors in Computing Systems 2026 (Position paper for "Herding CATs: Making Sense of Creative Activity Traces" workshop). [link]

[P4] ***DesignWeaver: Dimensional Scaffolding for Text-to-Image Product Design***
Sirui Tao, Ivan Liang, Cindy Peng, Zhiqing Wang, Srishti Palani, and Steven P. Dow.
Conference on Human Factors in Computing Systems 2025. [link]

[P3] ***HotSpot: Screened Poisson Equation for Signed Distance Function Optimization***
Zimo Wang, Cheng Wang, Taiki Yoshino, **Sirui Tao**, Ziyang Fu, and Tzu-Mao Li.
Conference on Computer Vision and Pattern Recognition 2025 (Highlight). [link]

[P2] ***Physion++: Evaluating Physical Scene Understanding with Objects Consisting of Different Physical Attributes in Humans and Machines***
Hsiao-Yu Tung, Mingyu Ding, Zhenfang Chen, **Sirui Tao**, Vedang Lad, Daniel Bear, Chuang Gan, Josh Tenenbaum, Daniel Yamins, Judith Fan, and Kevin Smith.
Proceedings of the Annual Meeting of the Cognitive Science Society 2023 (Poster with abstract). [link]

[P1] ***Physion: Evaluating Physical Prediction from Vision in Humans and Machines***
Daniel Bear, Elias Wang, Damian Mrowca, Felix Binder, Hsiao-Yu Tung, Pramod RT, Cameron Holdaway, **Sirui Tao**, Kevin Smith, Fan-Yun Sun, Fei-Fei Li, Nancy Kanwisher, Josh Tenenbaum, Dan Yamins, and Judith Fan.
Advances in Neural Information Processing Systems 2021 (NeurIPS Datasets & Benchmarks Track). [link]

Research Mentorship

Current Alisa Liao, B.S. UCSD [P5]
Current Shreya Krishnamurthy, B.S. UCSD [P5]
Alumni Cecilia Lin, B.S. UCSD [P5]
Alumni Kiruthika Marikumaran, B.S. + M.S. UCSD [P5]
Alumni Domonick Marshall, B.S. UCSD [P5]
Alumni Erin Huang, B.S. UCSD [P5]
Alumni Suma Vintha, B.S. UCSD [P5]

Research Experience

2023-12 - Present
UCSD, ProtoLab - Graduate Student Researcher (La Jolla, CA, USA)
Investigating generative models for contextual multimodal documents to aid designers and support collaborative scenarios.

2023-08 - 2025-04
UCSD, Tzu-Mao Li's Lab - Graduate Student Researcher (La Jolla, CA, USA)
Combining neural methods and traditional graphics pipelines to enable flexible and fast-forward physical interaction prediction; studying inverse physical prediction using diffusion-based models.

2022-11 - 2023-05
UCSD, HDSI - Undergraduate Researcher (La Jolla, CA, USA)
Developed GraphHSCN, a novel graph neural network architecture for long-range interaction modeling.

2021-01 - 2022-06
UCSD, CogTools Lab - Undergraduate Researcher (La Jolla, CA, USA)
Contributed to Physion and Physion++, focusing on data analysis and designing 3D stimuli for non-rigid physical interactions.

Industry Experience

2026-10 - Present
Smart Learning Hub - Designer
Designing useful, delightful experiences for instructors and students as a UCSD GSR. Built a collection of 124k+ course listings from 27 universities and 288 teaching-plan examples.

2026-06 - 2026-09
Autodesk - PhD Intern, AI Research – Datasets & Benchmarks (Remote)
Developed the Process–Artifact–Learning (PAL) evaluation protocol and a Fusion add-in that records AI-assisted CAD proposals, edits, validation checks, and human decisions. Primary mentor/manager: Will McCarthy; mentors: Vivian Liu and Anubrata Das.

2024-01 - 2024-04
MAVERiC Studio - Garage Startup Intern (La Jolla, CA, USA)
Explored virtual world creation, metaverse audience engagement, and AI tools. Built 'Not A Good Driver'. Try it here.

2023-01 - 2023-08
SchedGo - Data Engineer (Remote)
Built data infrastructure and conducted analytics; developed marketing data analytics to identify key user groups; designed a survey model for front-end user feedback.

2022-11 - 2023-08
San Diego Supercomputer Center - Deep Learning Engineer Intern (La Jolla, CA, USA)
Implemented scalable multi-GPU hyperparameter tuning using Ray Tune on HPC infrastructure. Investigated data-centric MLOps solutions to improve time and resource efficiency in experimentation.

2022-06 - 2022-09
Tesla - Data Scientist Intern (Fremont, California, USA)
Developed lossless image compression solutions and advanced statistical process control tools.

2021-07 - 2021-09
Buhler Group - Data Scientist Intern (Wuxi, Jiangsu, China)
Initiated MLOps image-labeling features for defect detection and prototyped cross-functional innovation tools.

2021-06 - 2021-07
J.D. Power - Analyst Intern (Shanghai, China)
Optimized workflow for quality analytics by automating data processing and report pipelines.

2020-08 - 2020-11
Bosch - Data Analyst Intern (Wuxi, Jiangsu, China)
Developed Tableau dashboards for manufacturing analytics; reduced bottleneck diagnosing time by 80%.

Volunteering

2021-09-01 - 2022-06-01
Halıcıoğlu Data Science Institute, UC San Diego - Data Science Student Representative (La Jolla, CA, USA)
Represented HDSI students to improve the data science program and curriculum.
- Peer-advised HDSI undergraduate students.
- Evaluated and improved the data science curriculum.
- Promoted the program to external audiences.

2021-09-01 - 2022-02-01
Mentor Collective - Mentor at UCSD (La Jolla, CA, USA)
Mentored first-year and transfer students at UCSD to help them adapt to university life.
- Provided guidance and support for new UCSD students.
- Facilitated connections to campus resources and communities.

Teaching Experience

2026-03 - 2026-06 **Teaching Assistant - COGS 125: Advanced Interaction Design** - UCSD
TA for Prof. Scott Klemmer; La Jolla, CA, USA.

2026-01 - 2026-03 **Teaching Assistant - DSGN 1: Design of Everyday Things** - UCSD
TA for Prof. Michael Meyer; La Jolla, CA, USA.

2025-08 - 2025-09 **Teaching Assistant - COGS 187A: Usability & Information Architecture** - UCSD
TA for Prof. Mary ET Boyle; La Jolla, CA, USA.

2025-08 - 2025-09 **Teaching Assistant - COGS 186: Genetic Algorithms** - UCSD
TA for Prof. Anjum Gupta; La Jolla, CA, USA.

2025-01 - 2025-03 **Teaching Assistant - DSGN 1: Design of Everyday Things** - UCSD
TA for Prof. Michael Meyer; La Jolla, CA, USA.

2024-09 - 2024-12 **Teaching Assistant - DSGN 1: Design of Everyday Things** - UCSD
TA for Prof. Michael Meyer; La Jolla, CA, USA.

2024-06 - 2024-09 **Teaching Assistant - CSE 175: Entrepreneurship for Engineers** - UCSD
TA for Prof. Rakesh Kumar; La Jolla, CA, USA.

2024-03 - 2024-06 **Teaching Assistant - DSGN 1: Design of Everyday Things** - UCSD
TA for Prof. Scott Klemmer; La Jolla, CA, USA.

2024-01 - 2024-03 **Teaching Assistant - COGS 9: Introduction to Data Science** - UCSD
TA for Prof. Meenakshi Khosla; La Jolla, CA, USA.

Academic Service

2026 IEEE Computer Graphics and Applications 2026 Journal Reviewer: 1 paper.

2026 UIST 2026 Full Paper Reviewer: 5 papers; Special Recognition for 2 + Highly Useful for 2.

2026 Human-Computer Interaction 2026 Journal Reviewer: 1 paper.

2026 CHI 2026 UCSD Happy Hour Party Co-organizer

2026 C&C Full Paper Reviewer: 2 papers; Special Recognition for 1.

2026 DIS Full Paper Reviewer: 2 papers; Special Recognition for 1.

2026 CHI Full Paper Reviewer: 8 papers; Special Recognition for 3 + Highly Useful for 1.

2025 - 2026 UCSD Design Lab Weekly Research Meeting Annual Organizer

2025 CI Extended Abstract Reviewer + Student Volunteer: 1 paper.

2025 CHI Late Breaking Work Reviewer: 2 papers.

2025 DIS Work-in-Progress Reviewer: 1 paper.

2024 UCSD Design Lab Weekly Research Meeting Fall Quarter Co-host

Projects

2026 **Website Revamp** - Portfolio redesign
Redesigned this portfolio as a small reflective-practice experiment in AI-assisted coding, visual critique, and research storytelling.

2024 **Not A Good Driver** - VRChat world
Built a playful VRChat world while exploring virtual world creation, audience engagement, and AI-assisted production workflows.

2023 **Context-Aware Encoding for LLMs** - LLM context prototype
Developed a tree-of-thoughts context encoding experiment to improve retrieval and contextual understanding.

2023 **GraphHSCN** - Graph neural network prototype
Developed a novel graph neural network architecture for long-range interaction modeling.

Honors & Awards (& Rejections)

2026	ChatGPT for Academic Researchers - OpenAI Lab access thanks to Steven :)
2026	Claude Team Plan for Scientists - Anthropic Lab access thanks to Steven :)
Jun 15, 2026	~~MSR AI and the New Future of Work CFP~~ - Microsoft Research **[Rejected Once, Accepted Zero Times ;)]**
Feb 27, 2026	~~MSR Fellowship~~ - Microsoft Research **[Rejected Once, Accepted Zero Times ;)]**
2026	YC AI Student Starter Pack - Y Combinator ~$25,000 in free credits for AI tools and cloud services.
2025 - 2031	SHORE Fellowship - UCSD, Cognitive Science Department ~$70,000 in value; 3 recipients in the Cognitive Science Department annually.
2025	OpenAI Researcher Access Program - OpenAI ~$1,000 in credits.
2023 - 2024	IGE Shah Fellowship - UCSD, Institute for the Global Entrepreneur Awarded the Shah Family Entrepreneur Fellowship ($5,000) as one of two Shah Fellows university-wide for outstanding research contributions and entrepreneurial potential. The fellowship, launched by the Institute for the Global Entrepreneur (IGE)—a collaboration between the Jacobs School of Engineering and the Rady School of Management—aims to teach engineers to become entrepreneurs and leaders.
2023	The Cornell, Maryland, Max Planck Pre-doctoral Research School Fellowship - Cornell, Maryland, Max Planck Institute $3,000; ~60 global recipients annually.
2021, 2022	HDSI Research Scholarship - UCSD, Halıcıoğlu Data Science Institute **[Rejected Once, Accepted Twice]** $2,500 x 2; ~30 recipients at HDSI annually.
2020 Spring, 2022 Winter	TRELS Research Scholarship - AEP, UCSD **[Rejected Twice, Accepted Twice]** $1,000 x 2; ~75 recipients at UCSD quarterly.
2019 - 2023	Provost Honor - Warren College, UCSD $0, but it sounds cool; ~20% of students at Warren College annually.

Skills

HCI	User-Centered Design, Interaction Design, Prototyping, Qualitative & Quantitative Methods, A/B Testing
Data Science	Scalable Analytics, Data Visualization, MLOps, Recommender Systems, Data Mining, Database
AI/ML	LSTM, GAN, Diffusion Models, GNN, NeRF, 3D Gaussian Splatting, Reinforcement Learning, LLM
Software	Python, R, SQL, Angular, D3.js, PyTorch, ClickHouse, FireStore, DevOps, Agile, AR/ VR
Mathematics	Probabilistic Reasoning, Descriptive and Inferential Statistics, Stochastic Processes

Languages

English (Fluent); Chinese (Native); German (Newbie)

Interests

Research: Design, HCI, Graphics, Creativity, Generative AI, 3D Understanding, Human & Embodied Autonomous Systems, Accessibility, Fairness

Real ones: Workout, Cats & Dogs, Cooking, Climbing, Nature, Movie, Game, Badminton, Travel

References

Professor Steven P. Dow - Master Advisor for my research in HCI.

Professor Tzu-Mao Li - Master Advisor for my research in Graphics.

Professor Judith E. Fan - Undergraduate advisor for my research in Intuitive Physics.

Note

I include all my rejections on my CV following the steps of the awesome Bradley Voytek (his CV). He said that would inspire more people to be ok with rejection, and I agree.